

Day 10 - Interview Questions

1. What is data persistence in Python?

Data persistence refers to the ability to store and retrieve data between program executions, ensuring that data remains available even after the program terminates.

2. How can you open and read a text file in Python?

You can open and read a text file in Python using the `open()` function and methods like `read()`, `readline()`, or `readlines()`.

3. What is the purpose of the with statement when working with files in Python?

The with statement is used to ensure that files are properly closed after they are used, preventing resource leaks.

4. Explain the difference between reading a file in text mode ('r') and binary mode ('rb').

Text mode reads and writes data as text (Unicode characters), while binary mode reads and writes data as raw binary bytes.

5. Explain how to read a file line by line in Python.

You can use a for loop to iterate through each line in a file by iterating over the file object itself.

6. How can you check if a file exists before attempting to open it in Python?

You can use the `os.path.exists()` function from the `os` module to check if a file exists before opening it.

7. What is the purpose of the strip() method when reading lines from a file?

The `strip()` method is used to remove leading and trailing whitespace (including newline characters) from lines read from a file.

8. What is the difference between reading a file using read() and readlines() methods in terms of memory usage?

`read()` reads the entire file into memory as a single string, which can be memory-intensive for large files. `readlines()` reads the file line by line and returns a list of lines, which can be more memory-efficient.

9. How can you read the first N lines of a file in Python without reading the entire file?

You can use a loop and the `readline()` method to read the first N lines from a file without reading the entire file.

10. How do you open a file for reading in Python, and what is the default mode when you don't specify it?

You can open a file for reading using the `open()` function with the `'r'` mode. If you don't specify a mode, `'r'` (read) is assumed by default.